Welfare Effects of Implicit Auction on Interconnector Capacity: Evidence from Japan

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ABSTRACT

This study examines the economic impact of implicit auction on the interconnector transmission capacity in Japan. The first-come, first-served rule allows incumbents to exercise vertical market power by withholding capacity from the day-ahead market. The implicit auction prevents them from reserving their physical transmission capacity and increases trade across zones in the day-ahead market. I use machine-learning methods to estimate the welfare impact of implicit auction. I predict the counterfactual market outcomes without implicit auction, and find that the trade effect of implicit auction is over $ 200 million per year. The study also finds that implicit auction does not necessarily reduce the price gap because it not only has a trade effect, but also a counteracting volume effect.

Keywords: *transmission right, market power, congestion management, interconnector, implicit auction*

1. INTRODUCTION

Transmission lines play a crucial role in an electric power network. They contribute to balancing demand and supply in real-time, guaranteeing efficient merit-order dispatch in wider geographic areas, and integrating intermittent renewable energy. In addition, transmission capacity increases competition in the generation and retail sectors by creating wider markets. However, interconnector transmission capacity is a scarce resource because, historically, each vertically integrated utility is required to supply all customers in its area under a regional monopoly and does not depend on imports across interconnectors. Hence, the efficient allocation of limited capacity is important, especially after market

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liberalization. Allocations are done through either market or non-market based congestion management methods (European Transmission System Operators 2004). What is the economic impact of market-based congestion management methods?

This study empirically evaluates the welfare impact of implicit auction relative to first-come, first-served (FCFS) allocations in Japan. Implicit auction simultaneously allocates interconnector capacity with energy in the day-ahead market, while the FCFS rule allows incumbents to reserve the capacity before the energy market. The FCFS rule effectively allocates it as a free non-tradable physical transmission right (PTR), which provides the right holders with an instrument to exercise vertical market power (Bushnell 1999). They can withhold PTR from the subsequent day-ahead market to create market-splitting and sell at higher market prices in the import constraint zone by intentionally overestimating their capacity needs. There is a penalty surcharge for canceling the reservation; however, the size of the surcharge is limited, and capacity holders can reduce the reservation even after the day-ahead market. Implicit auction prohibits interconnector capacity reservations in advance, and all capacity is released to the day-ahead market, thereby enabling more cross-zonal trade in the energy market. Increased trade reduces the price gap between the import and export zones. Implicit auction also improves the liquidity of the day-ahead market, because firms that previously reserved interconnector capacity for bilateral contracts outside the energy market are required to bid in the market.

I focus on two interconnectors where congestion was severe: Hokkaido-Honshu (HH) interconnectors connecting the Hokkaido and East zones, and Kanmon interconnectors connecting the West and Kyushu zones. The empirical challenge is that market outcomes are unobservable if FCFS has not been abolished. I use several machine-learning methods to predict counterfactual market outcomes in the post-treatment period. By comparing the loss functions, I evaluate the predictive performance and choose the best model to predict the counterfactuals. Moreover, by comparing actual and counterfactual market outcomes, I estimate the welfare impact of implicit auctions.

I find that the implicit auction improved the efficient allocation of interconnector capacity and prevent market power, relative to the FCFS rule. Furthermore, I find that the implicit auction has an economically significant impact on welfare. The trade effect is approximately $ 40 million for Hokkaido-Honshu interconnector and $ 60 million for the Kanmon interconnector over the six months of implementation. The annual gain is over $ 200 million/year.

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My contributions are three-fold. First, this study presents evidence of the impact of an implicit auction, compared to the non-market based FCFS rule. Numerous studies have examined the inefficiency of explicit auction, another market based method in comparison to implicit auction in Europe (e.g., Bunn and Zachmann 2010; Brunekreeft, Neuhoff, and Newberry 2005; Creti, Fumagalli, and Fumagalli 2010; Ehrenmann and Neuhoff 2009; Füss, Mahringer, and Prokopczuk 2020; Gugler, Haxhimusa, and Liebensteiner 2018; Keppler, Phan, and Le Pen 2016; Newbery, Strbac, and Viehoff 2016). Implicit auction can avoid the economically inefficient use of interconnector capacity, that is, export from high-price zones to low-price zones, attributable to the explicit auction. However, this study shows that implicit auction can also reduce the underutilization of interconnector transmission capacity caused by the cancellation of reserved interconnector capacity under FCFS. Moreover, I show that implicit auction has not only a price-decreasing effect (trade effect) on the day-ahead market price gap through releasing the interconnector capacity previously reserved under FCFS, but also a price-increasing effect (volume effect) by forcing those who reserved the capacity to participate in the market.

Second, I use empirical evidence to show that, the FCFS rule allowed PTR holders to exercise vertical market power, as predicted by the theoretical literature (e.g., Joskow and Tirole 2000; Bushnell 1999), and how an implicit auction could prevent this by simultaneously allocating the PTR in the day-ahead market.

Third, I avoid the potential omitted variable bias that can generally occur when studying two countries in Europe (e.g., Germany and France), which are connected by other neighboring countries and whose relevant data are often not available. By contrast, Japan is not connected to other countries, and all zonal data are available. An additional advantage is that its network is effectively radial and the analysis does not suffer from loop-flow issues.

The remainder of this paper is organized as follows. Section 2 describes the wholesale market structure, and section 3 explains interconnector congestion management methods in Japan. Section 4 defines the two effects of an implicit auction: trade and volume effects. Section 5 presents the research design, and Section 6 explains the data. Section 7 presents the estimates of the trade effect. Section 8 presents the counterfactual simulations and the welfare impact of implicit auctions and section 9 concludes the paper.

2. WHOLESALE MARKET STRUCTURE IN JAPAN

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Figure 1 shows that the Japanese day-ahead market comprises of nine fixed areas connected by interconnectors. Each area is equivalent to the control area in which the former vertically integrated monopoly operates. I classify the nine areas into four zones: Hokkaido, East, West, and Kyushu. Hokkaido is located in the northeastern part of Japan and has the highest day-ahead market price during the study period, partly because it still uses oil-fueled thermal power plants. The East zone includes Tokyo and Tohoku.

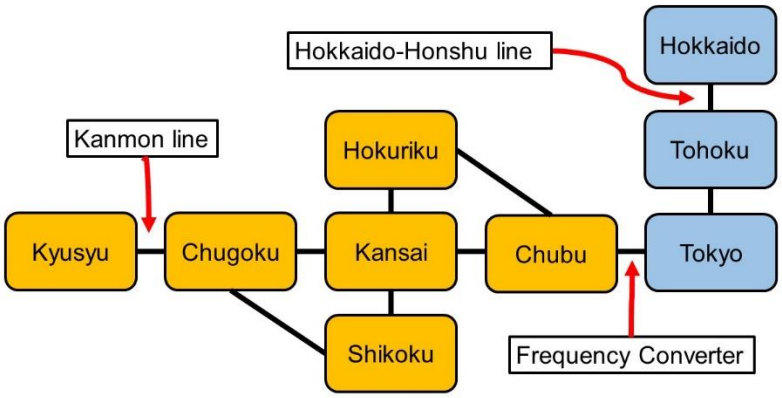


Figure 1. Day-ahead market price zones in Japan

The Hokkaido and East zones are connected by a high-voltage direct current HH interconnector. I defined the West zone as the compilation of Chubu, Kansai, Hokuriku, Shikoku, and Chugoku areas and regarded it as a single zone, as there is little congestion within the zone, and the market price is almost the same. The East and West zones are connected by high-voltage direct current frequency converters[2]. Kyushu is located in the southwestern part of Japan. The West and Kyushu zones are connected by a high-voltage alternate current Kanmon interconnector. These three interconnectors experienced frequent congestion. The remaining interconnectors are mostly uncongested during the study period. Although there is an intraday market, almost all energy is traded in the day-ahead market.

3. CONGESTION MANAGEMENT METHODS: FCFS AND IMPLICIT AUCTION

In Japan, the day-ahead market uses a single-price auction and a market-splitting method. The market-splitting algorithm allocates all net available interconnector capacity (NAIC) to

[2] Eastern and western Japan use 50 Hz and 60 Hz, respectively.

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market participants who successfully contract in the auction (Marmiroli et al. 2009). Before October 2018, the FCFS rule allowed incumbents to reserve their capacity for free *before* the start of the day-ahead auction at 10 am. Under FCFS, the NAIC in the day-ahead market is calculated as

NAIC = Operating capacity – Margin – Reserved capacity.

Operating capacity is the maximum amount of energy that can flow through an interconnector. The limit is the minimum value that satisfies all four constraints: thermal capacity, synchronization stability, voltage stability, and frequency. Margin is the reserved capacity to import or export energy from other areas to maintain system security during an emergency. The operating capacity and margin are calculated annually by the organization for cross-regional coordination of transmission operators (OCCTO), an association of regulators and transmission system operators. The operating capacity is stable except during planned maintenance. The margin is set to be constant for HH and zero for Kanmon. It should be noted that an implicit auction was already used in the market under the FCFS rule, but it can only be applied to the remaining NAIC.

FCFS based reservation is allowed for up to 10 years and could also be exclusively renewed. The reservation was effectively a non-tradable PTR (Joskow and Tirole 2000; Bushnell 1999; Gilbert, Neuhoff, and Newbery 2004). The market-split algorithm provides incumbents with an incentive to create congestion strategically and increase market revenue by selling at higher prices in an importing zone. The FCFS rule can give PTR holders the ability to exercise market power and avoid the risk of market price divergence between zones (Twomey et al. 2005). They can withhold interconnector capacity from the energy market by nominating it as a scheduled flow.

Under the FCFS rule, most interconnector capacities were used for bilateral contracts outside the wholesale energy market. Figure 2 shows the 70–100 TWh/year energy flow interconnectors for bilateral contracts, until 2017. This figure corresponds to approximately 7–10% of annual national power consumption. After the implicit auction was implemented in October 2018, the amount of trade via bilateral contracts dropped substantially to approximately 0.25 TWh/year. Meanwhile, the amount of energy traded in the day-ahead market skyrocketed, implying that the energy for bilateral contracts now flows into the day-ahead market.

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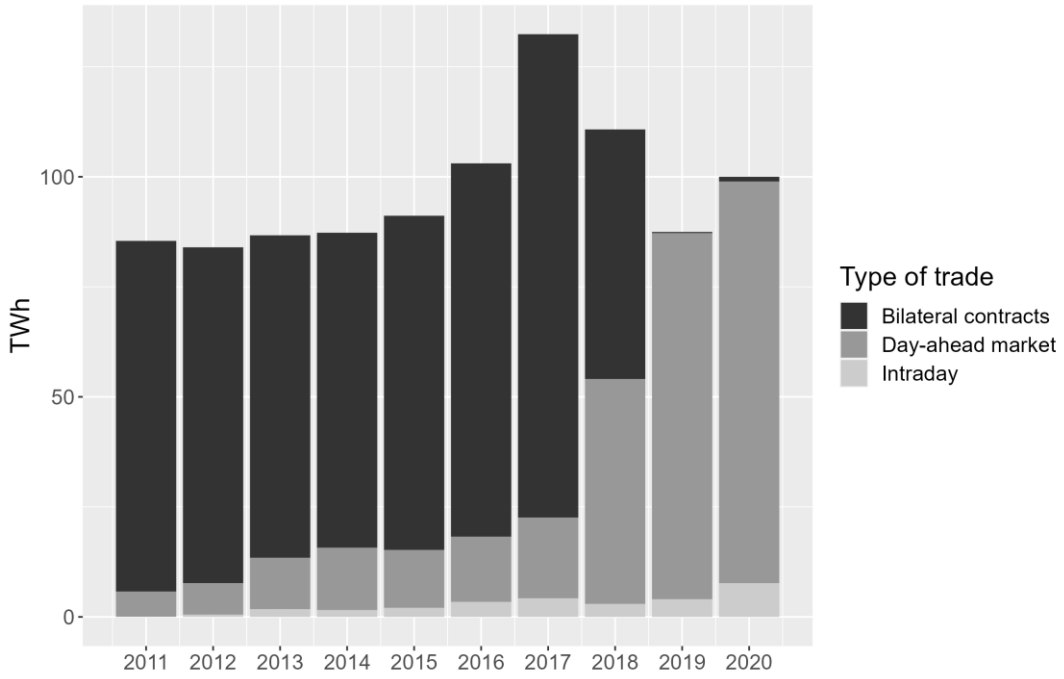
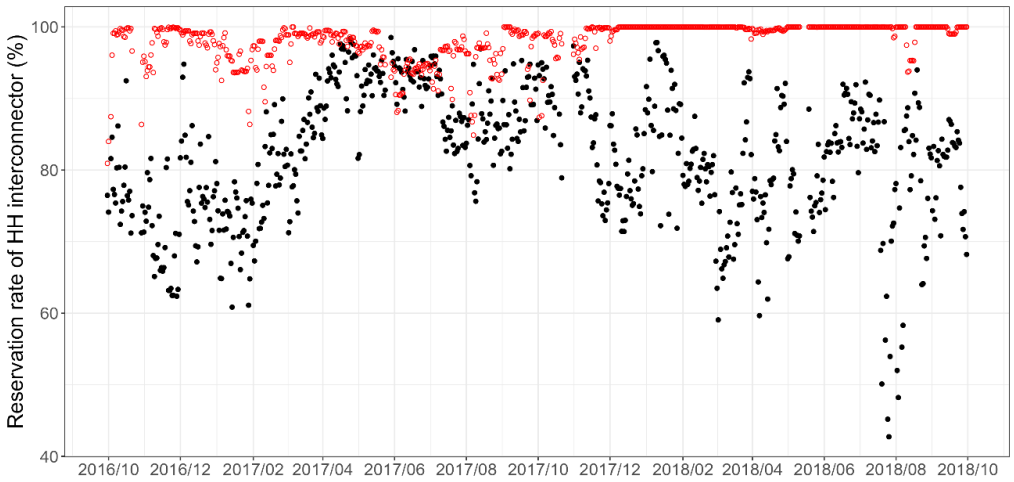


Figure 2. Cross-zonal trade volume via interconnectors by trading methods (TWh/ Fiscal year). Source: OCCTO

Figure 3 shows the daily average reservation rates for HH (Panel A) and Kanmon (Panel B) from October 2016 to September 2018. The red circle shows the reservation rate seven days before delivery, and the black dot is the reservation rate two days before delivery. The direction of the reserved flow is from East to Hokkaido in Panel A, and from Kyushu to West in Panel B. Both interconnectors are reserved in one direction. As expected, the reservation rate in both interconnectors is almost 100% seven days before delivery, implying that there is no available transmission capacity for the day-ahead market. While some of the capacity is released two days before delivery for HH, approximately 80% of capacity is already allocated before the day-ahead market. The difference between the reservation rate on the second and seventh day is very small in the case of Kanmon. This figure shows that the interconnector capacity available in the day-ahead market is limited for both interconnectors under the FCFS rule.

Panel A. HH

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Panel B. Kanmon

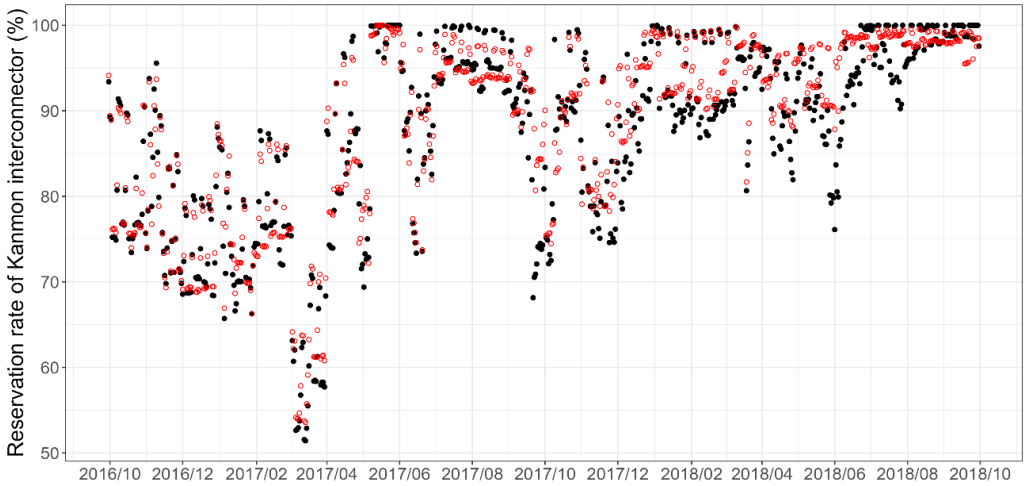


Figure 3. Daily average reservation rate under FCFS

Note: Red circles show the reservation rate seven days before delivery, and black dots are the reservation rate two days before delivery. Source: OCCTO.

Those who could reserve interconnector capacity could also cancel the reservation even after day-ahead market clearing, leaving an unused capacity. It should be noted that they could only cancel (i.e., reduce) the amount of reserved capacity, but could not increase it under FCFS. The cancellation does not mean exercising market power itself, but it implies that they could initially reserve more than they required, aggravating market power

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concerns. Those who canceled the reservation had to pay a penalty to prevent strategic overestimation of capacity needs. The surcharge was set at 0.01 yen/kWh of canceled capacity when they canceled more than 10% of the initial reserved capacity (OCCTO 2017). There was no surcharge for the capacity cancellation within 10% of the initial reservation. The low penalty implies that, they could still strategically withhold unnecessary interconnector capacity from the market. The right to cancel also helps the PTR holders to reduce the surplus imbalance.

Figure 4 shows how often reserved capacity was canceled after the day-ahead market. Panel A plots the relationship between the market-price gap between Hokkaido and the East zones in the day-ahead market and the reservation rate of the HH interconnector at 5 pm on the day before delivery. A negative reservation rate indicates that the interconnector capacity is reserved to transmit energy from East to Hokkaido. The two graphs on the left in each panel show the two-year observations from October 2016 to September 2018 under FCFS, and the two graphs on the right show the two-year observations after the full implicit auction was implemented from October 2018 to September 2020.

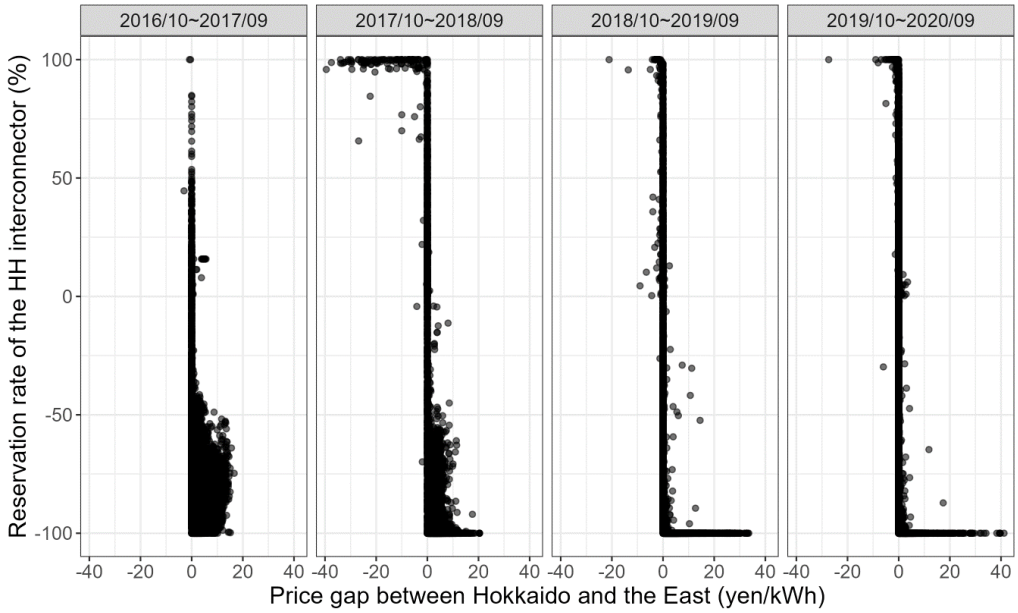
Under FCFS, the reservation rate of the HH interconnector was greater than 100% at 5 pm in numerous instances, even though the day-ahead market price in Hokkaido was higher than that in the East zone. The positive price gaps imply that congestion occurred at the HH interconnector during day-ahead market clearing, and all available capacity was allocated to the market, making a 100% reservation rate at 10 am. Hence, this figure confirms that the reserved interconnector capacity was frequently canceled after day-ahead market-splitting.

From October 2018, reservations before the day-ahead market were prohibited, and the NAIC was equal to the operating capacity minus the margin. Under a full implicit auction, incumbents could no longer reserve capacity, and all available capacity was allocated to the day-ahead market. The two graphs on the right show that most observations clustered at the dotted red lines after the implicit auction, indicating that they could neither reserve nor cancel the interconnector capacity.

Panel B of Figure 4 also shows that the market price in the West was higher than that in Kyushu for some hours, but the Kanmon interconnector capacity was not fully allocated at 5 pm. In sum, the FCFS rule allowed FTR holders to reduce reservation after the day-ahead market in both interconnectors, while implicit auction succeeded in fully allocating it in an economically efficient direction.

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Panel A. Price gap between Hokkaido and the East and the reservation rate of the HH interconnector



Panel B. Price gap between the West and Kyushu and the reservation rate of the Kanmon interconnector

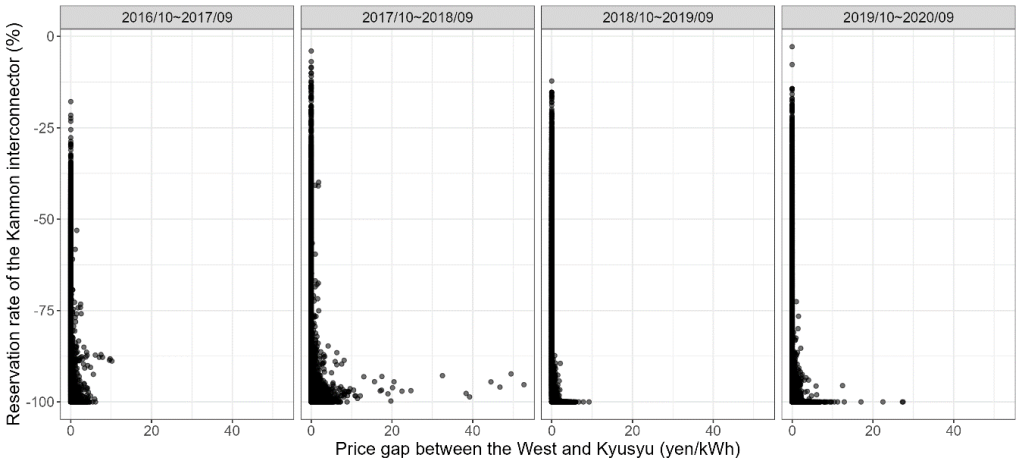


Figure 4. Market price gap and interconnector capacity reservation rate

Note: The two graphs on the left in each panel show the observations from October 2016 to September 2018, under the FCFS, and the two graphs on the right show the observations from October 2018 to September 2020 after the full implicit auction was implemented. A negative reservation rate indicates that the capacity is reserved for transmitting energy from the East zone to Hokkaido in Panel A and from Kyushu to the West zone in Panel B. Source: Japan Electric Power Exchange (JEPX) and OCCTO.

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4. EFFECTS OF IMPLICIT AUCTION

I argue that a full implicit auction (i.e., abolishing the FCFS rule) has two effects on the day-ahead market in Japan. The first is the trade effect. The implicit auction releases the entire interconnector capacity previously reserved for the energy market. This increases the NAIC and the trade quantity across market zones, similar to an interconnector capacity upgrade investment. Consequently, it will reduce the market price gap between the import and export zones. Figure 5 shows how an increase in the NAIC reduces the price gap between zones in the day-ahead market. $P^{export}_{Autarky}$ ($P^{import}_{Autarky}$) is the price in the export (import) zone under autarky. A price gap exists between $P^{export}_{Autarky}$ and $P^{import}_{Autarky}$. As the trade quantity increases, the price in the export zone increases to P^{export}_{Trade} and the price in zone B decreases to P^{import}_{Trade}, converging the price gap.

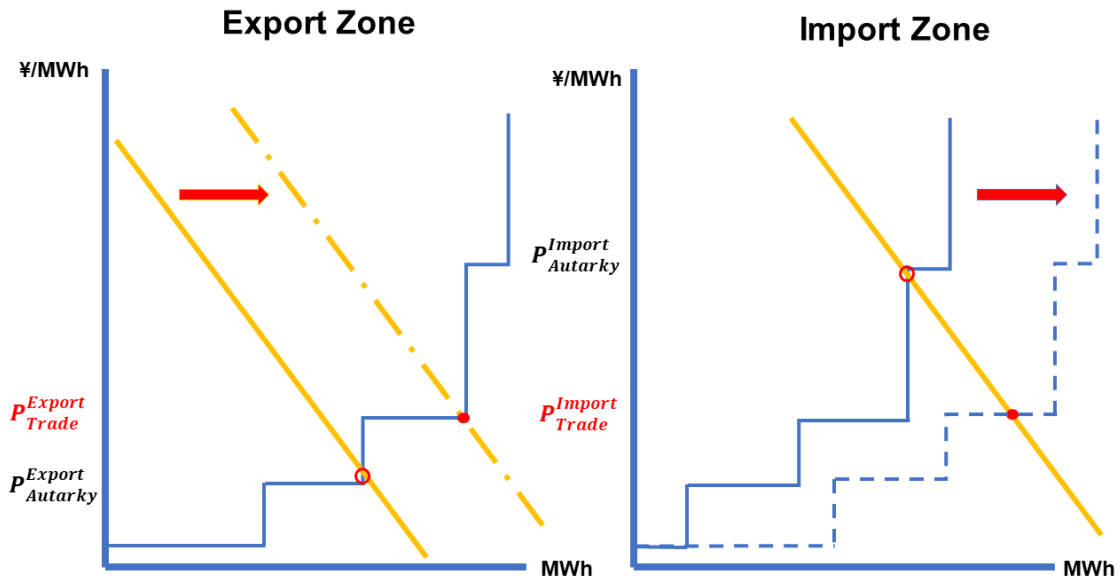


Figure 5. Trade Effect

The second is the volume effect. The full implicit auction prohibits reserving interconnector capacity for bilateral contracts across zones and thus increases the volume of

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the day-ahead market. After the full implicit auction was introduced in October 2018, those who used to reserve the interconnector capacity for bilateral contracts, were required to bid in the day-ahead market to obtain the interconnector capacity to trade energy across zones. Moreover, the regulator gave them temporary financial transmission rights to alleviate the unexpected risk of a price gap derived from market-splitting. This transitional measure allowed retailers who could successfully bid a day ahead to recover the price gap by March 2026. Thus, incumbents have a strong incentive to bid in the day-ahead market. Consequently, the full implicit auction increased the volume of both selling and buying quantities in the day-ahead market. Figure 6 shows both the volume of selling and buying increased discontinuously after October 2018.

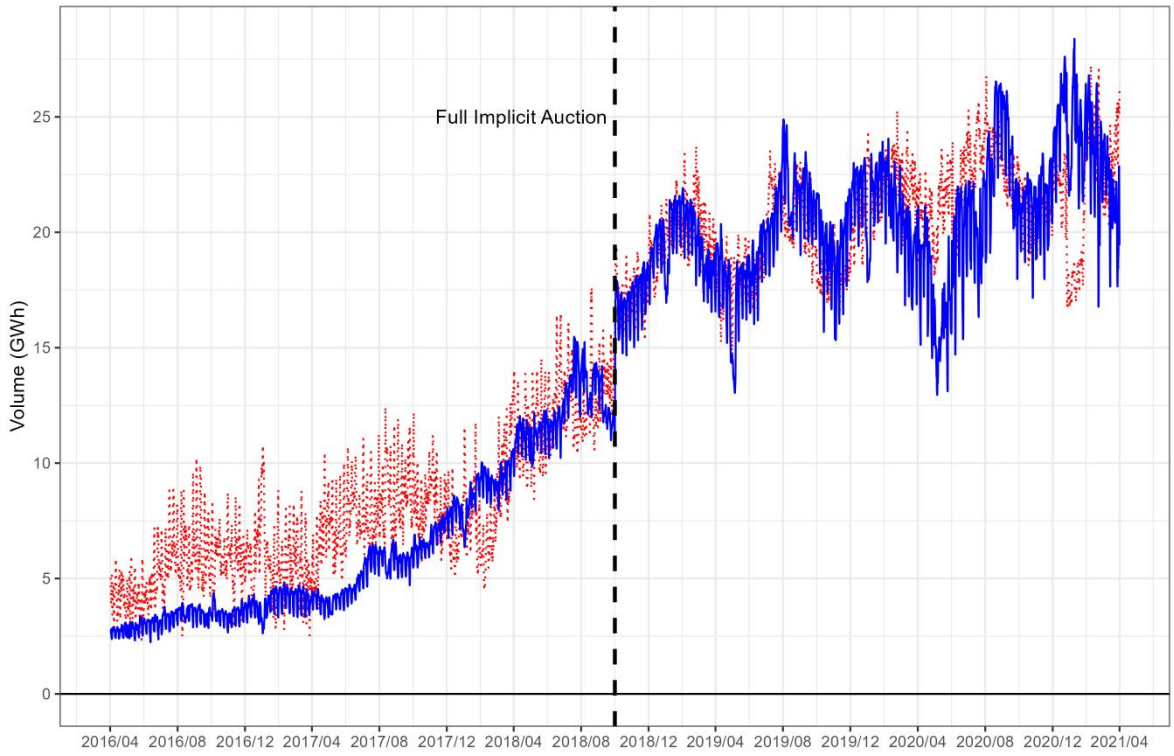


Figure 6. The volume of bidding and offer in the day-ahead market.

Note: The solid line is the daily moving average of the volume of selling, and the dotted line is the daily moving average of the volume of buying in the day-ahead market. Source: JEPX.

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This increase in the market volume may have affected the market price gap. When the additional offer price in the export zone is lower than the marginal cost of the marginal generator, the additional offer lowers the resulting zonal price. Otherwise, it does not affect the prices in the export zone. The left panel of Figure 7 depicts the former scenario. Similarly, if the additional bid price in the import zone is higher than the former equilibrium price, the addition raises the market price in the import zone. Otherwise, it does not affect the price in the import zone. The right panel of Figure 7 depicts the former scenario. In sum, the volume effect can widen the price gap between the import and export zones.

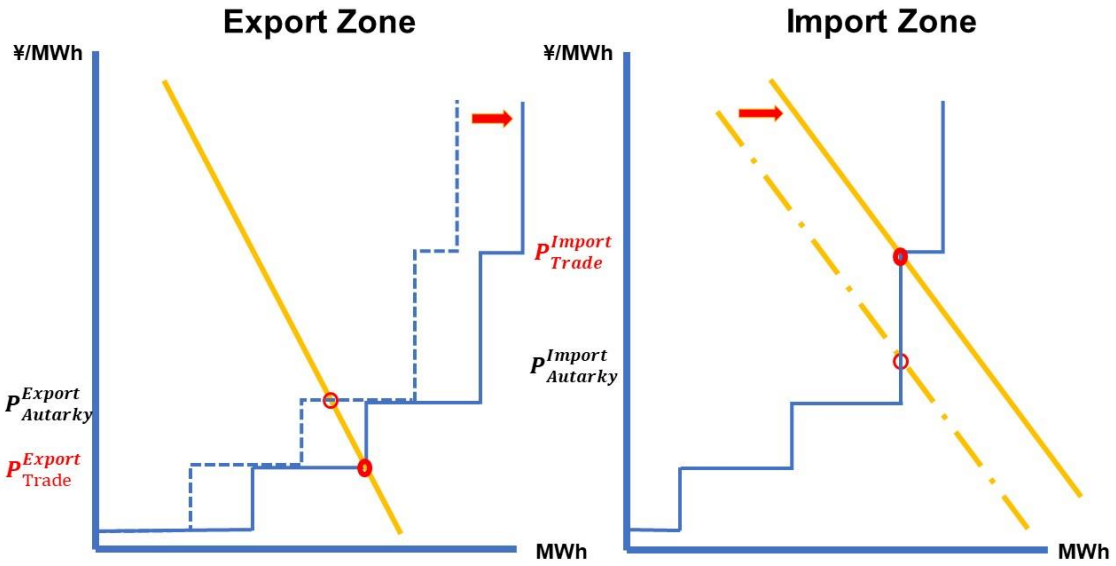


Figure 7. Volume effect

5. EMPIRICAL STRATEGY

To estimate the trade effect, I model the day-ahead market price gap between the import and export zones as a function of the NAIC, volume added to the market after the implicit auction, supply controls, demand controls, and fixed effects. The following equation is used:

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$$\Delta p_{DA,t} = \alpha_0 + \beta_1 NAIC_t + \beta_2 Volume_t + \beta_3 Supply_t + \beta_4 Demand_t + FE_t + \varepsilon_t \quad (1).$$

$\Delta p_{\mathrm{DA,t}}$ is the day-ahead market price difference between the import and export zones (Hokkaido and East for HH and West and Kyushu for Kanmon). NAIC is the interconnector capacity available in the day-ahead market. The NAIC has two different values depending on the flow direction. I use the NAIC from the export zone to the import zone. β_1 is the parameter used to measure the size of the trade effect.

Volume is the bid quantity added to the market after an implicit auction. As this variable is not observable, I use the amount of counterfactual reserve capacity as a proxy, assuming that the added bid quantity would be equal to the volume of counterfactual reserved capacity if the FCFS rule was maintained. This assumption seems plausible because those who tended to reserve capacity were not only required to bid in the market to obtain it but were also given the transitional financial transmission right conditional on making a successful bid in the market, as discussed in the previous section.

I include the supply and demand shocks that may affect the outcome and NAIC. Supply control variables include photovoltaic (PV) and wind generation in each zone. Nuclear power generation is supplied as a baseload and included as an exogenous variable. To control fossil fuel costs, daily coal prices, monthly liquefied natural gas (LNG) import prices, and daily Brent oil prices are used. Demand variables are the actual hourly electricity consumption in each zone. These are proxies for forecasted demand and are assumed to be exogenous in the short term.

FE_t includes hour, day of the week, month, and fiscal year fixed effects. I also include a month-year fixed effect to control for unobservable progress of the retail competition and the "gross bidding" policy, which started in April 2017 to encourage the incumbent generation or retail firms to voluntarily bid some of their internal contracts into the day-ahead market. Hour-month fixed effects are also added to control for prediction errors in renewable energy generation and demand. Standard errors are clustered by day to address autocorrelation.

6. DATA

The JEPX publishes half-hourly, day-ahead market price data for each area. I calculate the average hourly market prices (yen/kWh) to merge with the hourly covariates. The data

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on operating capacity, margin, and reserved interconnector capacity are available at OCCTO. I use the reserved interconnector capacity at 3 pm, two days before delivery, as the reserved interconnector capacity under the FCFS rule. This implicitly assumes that incumbents could have canceled the reservation *after* the day-ahead market on day t-1, but did not cancel from 3 pm on day t-2 to 10 am on day t-1. This assumption seems plausible because they had the incentive to withhold the capacity at least until the day-ahead market clearing to strategically create congestion and market splits. The volume variable is the predicted counterfactual reserved capacity, which is estimated in Section 8.

Supply and demand variables are gathered from transmission system operators' websites. The daily coal price data are obtained from the globalCOAL NEWC Index, while the daily Brent oil price is assembled by the Energy Information Administration. The monthly LNG import price is calculated based on Trade Statistics of Japan in the Ministry of Finance. The study period was from midnight on June 4, 2016 to 3 pm on March 28, 2019. I exclude the data afterward because the operating capacity of the HH interconnector was upgraded from 600 MW to 900 MW at 3 pm on March 28, 2019, and the financial transmission rights market was introduced on April 1, 2019, which may have changed the bidding behavior of the market participants.

Table 1 presents the descriptive statistics. The NAIC of the Kanmon interconnector is much larger than that of the HH interconnector. The Kanmon interconnector has an operating capacity of 2800 MW, whereas the HH interconnector has an operating capacity of 600 MW. Solar power generation in Japan is much greater than wind power generation. Nuclear power plants were present in all four zones but stopped operating after the nuclear accident in Fukushima in 2011. Only several nuclear power plants in the West and Kyushu zones were operational during the study period.

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Table 1. Descriptive Statistics (06/04/2016~03/28/2019)

Variable	Unit	Obs	Mean	Std. Dev.	Min	Max
Price gap between Hokkaido and East zones	yen/kWh	24,183	3.290603	4.370945	-39.51	39.115
Price gap between the West and Kyushu zones	yen/kWh	24,663	0.28956	1.160589	0	52.68
NAIC of HH	MW	24,663	46.67206	47.14436	0	200.496
NAIC of Kanmon	MW	24,663	710.1228	825.4094	0	2980
Volume for HH	MW	24,663	19.20276	42.91469	0	149.5973
Volume for Kanmon	MW	24,663	332.1523	726.4957	0	2301.543
Solar in Hokkaido	MWh	24,663	157.3206	236.4576	0	1103
Solar in the East	MWh	24,663	2016.408	3078.68	0	13901
Solar in the West	MWh	24,663	2481.806	3656.362	0	16161
Solar in Kyushu	MWh	24,663	1000.39	1532.196	0	6656
Wind in Hokkaido	MWh	24,663	99.80189	70.19188	0	317
Wind in the East	MWh	24,663	368.4876	281.3567	0	1338
Wind in the West	MWh	24,663	224.6692	165.2895	3	832
Wind in Kyushu	MWh	24,663	63.0356	58.8241	0	316
Nuclear in the West zone	MWh	24,663	2230.038	1463.747	0	5025
Nuclear in Kyushu	MWh	24,663	2127.796	1136.346	634	4151
Demand in Hokkaido	MWh	24,663	3568.859	635.8973	0	5422
Demand in the East	MWh	24,663	42478.48	7755.905	26482	69346
Demand in the West	MWh	24,663	46223.74	8358.808	28290	75471
Demand in Kyushu	MWh	24,663	10029.18	1706.838	6453	16011
Coal price	$/ton	24,663	93.44434	15.71042	50.46	122.89
LNG price	1000 yen/ton	24,663	49.16741	9.223131	32.87843	65.38621
Oil price	$/barrel	24,663	59.56472	10.92182	40	86.07

7. RESULTS

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Table 2 presents the estimation results of Equation (1). Column (1) is the baseline estimation with the hour, week, month, and fiscal year fixed effects. This indicates that a 1 MW increase in the NAIC of the HH line reduces the day-ahead market price gap between Hokkaido and the East zone by 0.014 yen/kWh (14 yen/MWh). The other control variables have expected signs and sizes. The volume variable is positively correlated with the outcome. Solar power generation in Hokkaido significantly reduces the market price gap by lowering the market price in Hokkaido. Wind generation remains mostly insignificant, partly because there is less variation than that in solar generation. Demand in Hokkaido is positively associated with the outcome, whereas demand in the East is negatively associated with the outcome. LNG prices are negatively associated with the market price gap. It is notable that, there was no LNG-fired power plant in Hokkaido during the study period; thus, the LNG price did not affect the market price in Hokkaido. Meanwhile, the rise in the LNG price increases the market price in the East and consequently reduces the price gap.

Columns (2) and (4) introduce additional fixed effects. Column (2) includes month×year fixed effects to flexibly control for the unobservable supply or demand shocks that may differ by month in the sample period. Column (3) includes hour×month fixed effects. Column (4) includes both month×year and hour×month fixed effects and is our preferred specification.

Table 2. Trade effect of the HH interconnector

Variable	1	2	3	4
NAIC	-0.014**	-0.024***	-0.015**	-0.026***
	[0.006]	[0.007]	[0.007]	[0.007]
Volume	0.042***	0.080**	0.042***	0.070**
	[0.013]	[0.030]	[0.013]	[0.027]
Solar in Hokkaido	-0.004***	-0.004***	-0.002***	-0.002***
	[0.001]	[0.001]	[0.001]	[0.001]
Solar in the East	0	0.000*	0	0
	[0.000]	[0.000]	[0.000]	[0.000]
Solar in the West	0	0	0	0
	[0.000]	[0.000]	[0.000]	[0.000]
Solar in Kyushu	-0.000**	-0.000**	0	0

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	[0.000]	[0.000]	[0.000]	[0.000]
Wind in Hokkaido	0	0	-0.001	0
	[0.001]	[0.001]	[0.001]	[0.001]
Wind in the East	0	0	0	0
	[0.000]	[0.000]	[0.000]	[0.000]
Wind in the West	0	0	0	0
	[0.001]	[0.001]	[0.001]	[0.001]
Wind in Kyushu	0	-0.001	0.001	0
	[0.001]	[0.001]	[0.001]	[0.001]
Nuclear in the West	0	0	0	0
	[0.000]	[0.000]	[0.000]	[0.000]
Nuclear in Kyushu	-0.001***	0	-0.001***	0
	[0.000]	[0.000]	[0.000]	[0.000]
Demand in Hokkaido	0.001***	0.002***	0.002***	0.002***
	[0.000]	[0.000]	[0.001]	[0.000]
Demand in the East	-0.000***	-0.000***	-0.000***	-0.000***
	[0.000]	[0.000]	[0.000]	[0.000]
Demand in the West	0	0.000**	0	0
	[0.000]	[0.000]	[0.000]	[0.000]
Demand in Kyushu	-0.000***	-0.001***	0	0
	[0.000]	[0.000]	[0.000]	[0.000]
Coal price	-0.018	0.044***	-0.022*	0.045***
	[0.011]	[0.013]	[0.011]	[0.013]
LNG price	-0.192***	-4.834***	-0.172***	-5.157***
	[0.031]	[1.568]	[0.030]	[1.522]
Oil price	-0.082***	-0.045	-0.078***	-0.036
	[0.021]	[0.028]	[0.021]	[0.029]
Month×Year FE	No	Yes	No	Yes
Hour×Month FE	No	No	Yes	Yes
Adj-R-squared	0.285	0.355	0.312	0.38

Day-cluster robust standard errors are shown in parentheses. All the specifications include hour, week, month, and fiscal year fixed effects. The total number of observations are 24,183. *** p<0.01, ** p<0.05, * p<0.1

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Table 3 shows the estimation results for the Kanmon interconnector across the West and Kyushu zones. A 1 MW increase in the NAIC of the Kanmon interconnector decreases the market price gap by 0.0004 yen/kWh (0.4 yen/MWh) between the West and Kyushu zones across all specifications. Column (4) is my preferred specification and is used for the welfare analysis in the next section.

Table 3. Trade effect of Kanmon interconnector

Variable	1	2	3	4
NAIC	-0.0004***	-0.0004***	-0.0004***	-0.0004***
	[0.0001]	[0.0001]	[0.0001]	[0.0001]
Volume	0.0006***	-0.0012**	0.0006***	-0.0012**
	[0.0002]	[0.0005]	[0.0002]	[0.0005]
PV in Hokkaido	0.0002	0.0003	0.0003	0.0003
	[0.0002]	[0.0002]	[0.0002]	[0.0002]
PV in the East	-0.0000*	-0.0000**	-0.0000**	-0.0000**
	[0.0000]	[0.0000]	[0.0000]	[0.0000]
PV in the West	0	0	0	0
	[0.0000]	[0.0000]	[0.0000]	[0.0000]
PV in Kyushu	0.0001***	0.0001***	0.0001***	0.0001***
	[0.0000]	[0.0000]	[0.0000]	[0.0000]
Wind in Hokkaido	-0.0004	-0.0005	-0.0004	-0.0005
	[0.0004]	[0.0004]	[0.0004]	[0.0004]
Wind in the East	0	0.0001	0	0.0001
	[0.0001]	[0.0001]	[0.0001]	[0.0001]
Wind in the West	0.0001	0	0.0001	0
	[0.0001]	[0.0001]	[0.0001]	[0.0001]
Wind in Kyushu	-0.0009***	-0.0008***	-0.0009***	-0.0008***
	[0.0002]	[0.0002]	[0.0002]	[0.0002]
Nuclear in the West	-0.0003***	-0.0001*	-0.0003***	-0.0001*
	[0.0001]	[0.0001]	[0.0001]	[0.0001]
Nuclear in Kyushu	0.0002***	0.0002**	0.0002***	0.0002**
	[0.0001]	[0.0001]	[0.0001]	[0.0001]

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Demand in Hokkaido	-0.0002**	-0.0002**	-0.0001*	-0.0001**
	[0.0001]	[0.0001]	[0.0001]	[0.0001]
Demand in the East	0.0000**	0	0.0000**	0
	[0.0000]	[0.0000]	[0.0000]	[0.0000]
Demand in the West	0.0000*	0.0000**	0.0000*	0.0000*
	[0.0000]	[0.0000]	[0.0000]	[0.0000]
Demand in Kyushu	-0.0001**	-0.0001	-0.0001	0
	[0.0000]	[0.0000]	[0.0000]	[0.0001]
Coal price	0.0121***	0.0079***	0.0121***	0.0079***
	[0.0026]	[0.0023]	[0.0026]	[0.0023]
LNG price	0.0262***	0.5888*	0.0273***	0.5836*
	[0.0085]	[0.2972]	[0.0087]	[0.3000]
Oil price	0.0095*	-0.0200**	0.0096*	-0.0194**
	[0.0054]	[0.0089]	[0.0055]	[0.0090]
Month×Year FE	No	Yes	No	Yes
Hour×Month FE	No	No	Yes	Yes
Adj-R-squared	0.1344	0.1636	0.1417	0.1712

Day-cluster robust standard errors are shown in parentheses. All the specifications include hour, week, month, and fiscal year fixed effects. The total number of observations are 24,663. *** p<0.01, ** p<0.05, * p<0.1

8. WELFARE ANALYSIS

8.1 Welfare effect of implicit auction

Figure 8 illustrates the welfare impact of implicit auction. Trapezium ABCD is the welfare gain from trade in the day-ahead market under implicit auction relative to autarky. The dotted and colored trapezium EFGH is the welfare gain from trade if the FCFS is maintained relative to autarky. $\hat{q}_{FCFS}$ is the counterfactual trade quantity under FCFS during the post-treatment period. The welfare impact of the implicit auction is equal to the difference between the area of trapezium ABCD and that of trapezium EFGH. The implicit auction increases the trade quantity q_{IA} relative to the counterfactual trade quantity $\hat{q}_{FCFS}$ under FCFS. This translates into greater economic welfare. The difference between $\Delta\hat{p}_{IA}^{Autarky}$ and $\Delta\hat{p}_{FCFS}^{Autarky}$ reflects the volume effect of the implicit auction; it can

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increase the price gap by increasing the amount of the bid/offer even in the absence of trade.

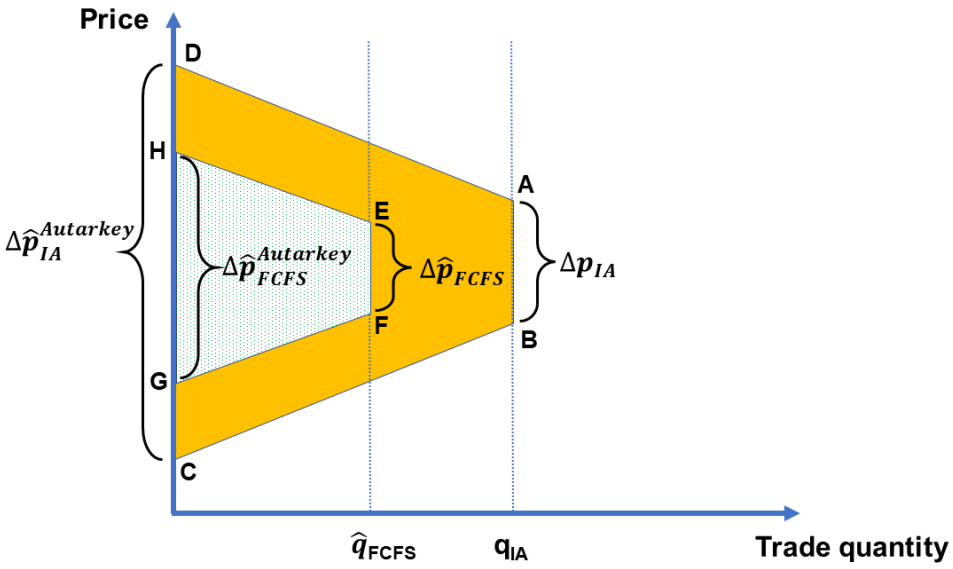


Figure 8. Welfare effect of implicit auction

Welfare gain of implicit auction relative to the FCFS rule is calculated as

$$\Delta W = \frac{1}{2}\left(\Delta p_{IA} + \Delta\hat{p}_{IA}^{Autarky}\right) \times q_{IA} - \frac{1}{2}\left(\Delta\hat{p}_{FCFS} + \Delta\hat{p}_{FCFS}^{Autarky}\right) \times \hat{q}_{FCFS} \qquad (2).$$

This is interpreted as production cost savings resulting from more efficient merit order dispatch across zones (Mansur and White 2012). Moreover, it prevents PTR holders from withholding interconnector capacity from the day-ahead market, which potentially comes from exercising market power.

8.2 Counterfactual prediction

To estimate the welfare gain of the implicit auction, I predict a counterfactual market price gap $\Delta\hat{p}_{IA}^{Autarky}$, $\Delta\hat{p}_{FCFS}$, $\Delta\hat{p}_{FCFS}^{Autarky}$, and trade quantity $\hat{q}_{FCFS}$ if the full implicit auction had not replaced FCFS after October 2018. To obtain the best predictive model, I

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use the least absolute shrinkage and selection operator (LASSO), random forest (RF), deep neural network (DNN), and linear regression to compare the mean square error (MSE) of the predicted counterfactuals. These machine-learning methods can flexibly approximate the conditional mean of the dependent variable, and predict counterfactuals better than linear regression.

LASSO adds a regularizer of the sum of the absolute values of the coefficients to the linear regression so that the model can avoid in-sample overfitting (Mullainathan and Spiess 2017). I use 10-fold cross-validation to select the optimal value of the tuning parameter. To train the model, I use the R package "gmnnet."

RF combines a regression tree model with bagging. The regression tree sequentially splits the sample based on the threshold value of an explanatory variable and predicts an outcome variable by splitting sub-samples. The RF takes the average of several hundred different trees constructed by bootstrapping the training sample with randomly chosen subsets of explanatory variables. One of the advantages of RF is that it requires relatively little tuning (Athey and Imbens 2019). To train the model, I use the R package "randomForest."

For LASSO and RF, I divide the pretreatment sample into training and test data in a ratio of eight to two. The sample is not randomly split to ensure that the training data are always older than the test data. Pretreatment data are collected from June 4, 2016, to September 30, 2019. I use the training data to estimate the model by minimizing the MSE. The test data are used to evaluate the predictive performance of the models with the loss function (MSE).

As DNN requires hyperparameter tuning, the sample is divided into training, validation, and test data in a ratio of six to two to two, respectively. Again, the sample is split nonrandomly. The validation data are used to select the optimal hyperparameters to avoid overfitting. The test data are reserved for evaluating the model performance. The training data are standardized by extracting the mean and dividing it by the standard deviation. The validation and test data use the same mean and standard deviation values for standardization. A deep neural network is trained using training data with a rectified linear unit activation function and the Adam optimizer, which is a stochastic gradient descent algorithm. Hyperparameters include several hidden layers, number of units, dropout rate, and learning rate. To find the best set of hyperparameters efficiently, a "hyperband" is used (Li et al. 2018). The number of epochs is set to 800, and the batch size is 64. To avoid overfitting, early stopping is introduced, which completes training when the MSE does not

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decline consecutively over five epochs. The model is trained using the "TensorFlow" and "Keras" libraries in Python. Table 4 lists the combinations of the search space of the hyperparameters and selected values.

Table 4. Hyperparameters for tuning

Hyperparameter	**Search space**
Number of hidden layers	1, 2, 3
Number of units	32, 64, 96, 128
Dropout rate	0, 0.25, 0.5
Learning rate	0.0001 to 0.001 (log sampling)

To calculate these counterfactuals, I first predict the counterfactual reserved interconnector capacity if the FCFS was maintained. I model the reserved interconnector capacity under the FCFS rule as a function of the gross available interconnector capacity (GAIC) and time-fixed effects:

$$q_{reserve,t} = \alpha_0 + \beta_1 GAIC_t + FE_t + e_t \qquad (3),$$

where $q_{reserve,t}$ is the reserved interconnector capacity under FCFS. G*AIC* is operating capacity minus the margin. This equation implies that the reserved interconnector capacity is restricted by G*AIC*. FE_t denotes the hour, week, and month effects.

Table 5 summarizes the predictive performance of each model. The first column is the result of the linear regression, second column, the LASSO, third column uses RF, and fourth column, the DNN. As the RF model has the smallest test MSE for both interconnectors, I predict the counterfactual reserved interconnector capacity if the implicit auction was not introduced by RF.

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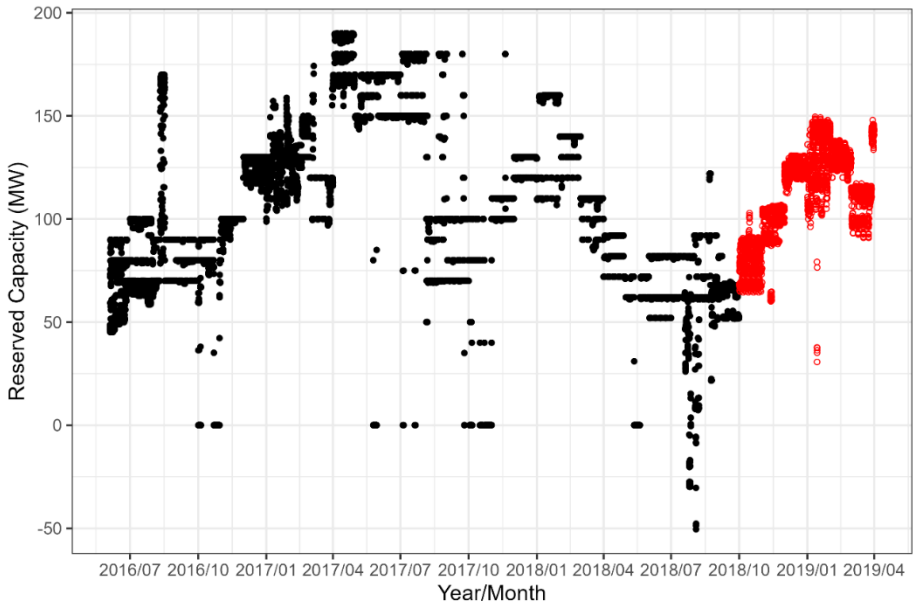
Table 5. Test MSE of reserved interconnector capacity

Model	(1) OLS	(2) LASSO	(3) RF	(4) DNN
HH	509.80	512.48	450.57	668.65
Kanmon	147977.1	147337.4	130437.5	229355.3

Note: In the RF, six variables are randomly sampled as candidates for each split.

Based on the parameters estimated by RF, I predict a counterfactual reserved interconnector capacity of HH and Kanmon, if the FCFS rule was maintained after October 2018, using the data in the post-treatment period: $\hat{q}^{post}_{reserve,t}$. Figure 9 shows the observed and counterfactual reserved interconnector capacity. The black dots indicate the actual reserved capacity before the implicit auction, and the red dots show the counterfactual reserved capacity after October 2018.

Panel A. HH



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Panel B. Kanmon



Figure 9. Reserved interconnector capacity under FCFS

Next, I predict the counterfactual price gap if the FCFS was not replaced by an implicit auction, as follows:

$$\Delta p_{DA,t} = \alpha_0 + \beta_1 NAIC_t + \beta_2 Supply_t + \beta_3 Demand_t + FE_t + \varepsilon_t \quad (4).$$

Equation (4) is the same as Equation (1), except that there is no volume variable in the model, and I only use data from the pretreatment period (i.e., before October 2018). Table 6 shows that, RF can predict the day-ahead market price gap between Hokkaido and the East with the lowest MSE, while OLS produces the lowest MSE for the West and Kyushu compared to other machine-learning methods.

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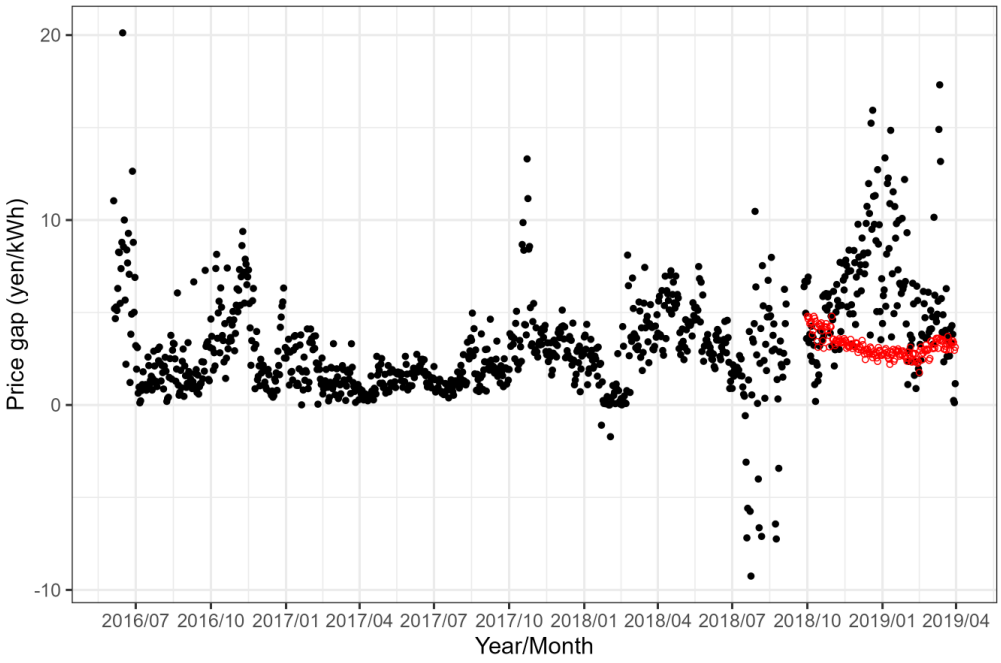
Table 6. Test MSE of day-ahead market price gaps

Model	(1) OLS	(2) LASSO	(3) RF	(4) DNN
Spread between Hokkaido and the East	21.29	21.50	19.55	20.70
Spread between the West and Kyushu	5.28	5.34	5.29	5.41

Note: In the RF, seven variables are randomly sampled as candidates at each split.

Thus, I use RF to predict the counterfactual price gap between Hokkaido and the East, and OLS to predict that of the West and Kyushu in the post-treatment period. Figure 10 presents the observed and counterfactual market price gaps. The black dots show the daily-mean actual price gap, and the circles represent the daily-mean counterfactual price gap, after October 2018.

Panel A. Counterfactual price gap between Hokkaido and the East



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Panel B. Counterfactual price gap between West and Kyushu

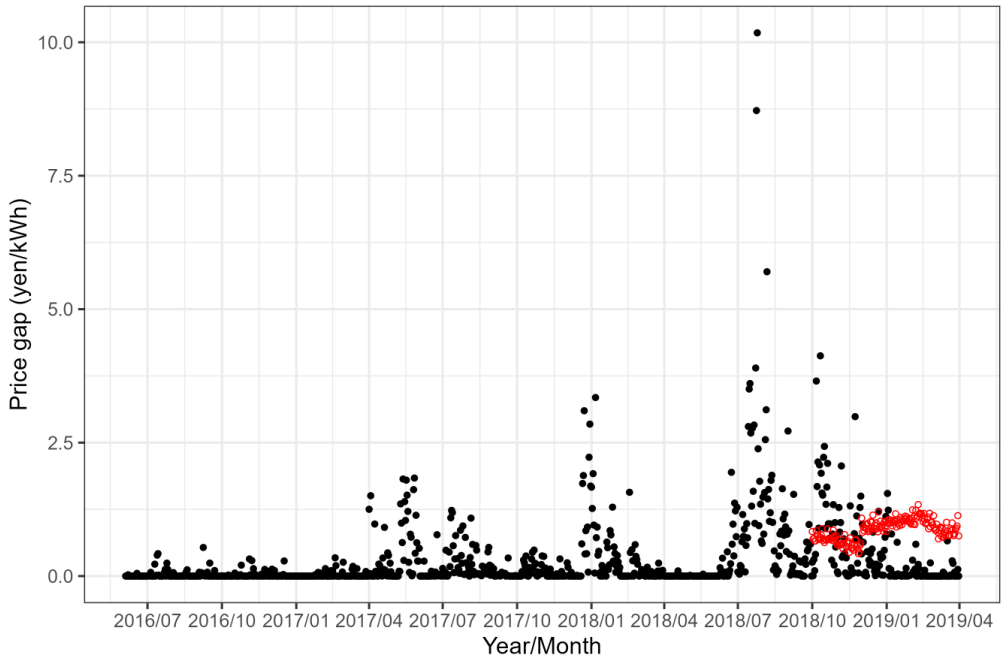


Figure 10. Day-ahead market price spread

I then estimate the counterfactual trade quantity $\hat{q}_{FCFS}$ allocated in the day-ahead market if the FCFS rule was not abolished as follows:

$$\hat{q}_{FCFS} = \widehat{NAIC}_t \quad if \quad \Delta\hat{p}_{DA,t} \neq 0 \qquad (5)$$

Note that the estimated $\widehat{NAIC}_t$ and counterfactual price gap $\Delta\hat{p}_{DA,t}$ are used. $\hat{q}_{FCFS}$ is equal to $\widehat{NAIC}_t$ as long as the market price gap is non-zero. This relationship holds because the market-splitting algorithm allocates all the remaining capacity in the wake of congestion (Marmiroli et al. 2009).

Next, I calculate the counterfactual day-ahead market price under autarky $\Delta p_{DA,t}^{autarky}$ as follows:

$$\Delta p_{DA,t}^{autarky} = \begin{cases} \Delta p_{DA,t} - \widehat{\beta_1} NAIC_t & under\ full\ implicit\ auction \\ \Delta\hat{p}_{\mathrm{DA,t}} - \widehat{\beta_1}\widehat{NAIC}_t & under\ FCFS \end{cases}$$

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$\hat{p}_{DA,t}$ is the estimated market price gap under FCFS. $\widehat{\beta_1}$ is the estimated coefficient of trade effect by Equation (1). $\widehat{NAIC}$ is predicted counterfactual, if the FCFS was maintained in the post-treatment period.

8.3 Welfare Analysis

Finally, a welfare analysis is conducted according to Equation (2). Table 7 summarizes the welfare effects of a full implicit auction. The estimated average price gap between Hokkaido and the East under autarky is 3.93 yen/kWh between April and September 2018, and 9.52 yen/kWh between October 2018 and March 2019. The estimated average price gap between the West and Kyushu under autarky is 0.96 yen/kWh before the full implicit auction, and 1.39 yen/kWh after October 2018. I estimate that the counterfactual average price gap between Hokkaido and the East zones under the FCFS after October 2018 would be 3.19 yen/kWh, compared with 3.11 yen/kWh before October 2018. I also find that the counterfactual average price gap between the West and Kyushu zones under the FCFS after October 2018 would be 0.86 yen/kWh for Kanmon, compared to 0.63 yen/kWh before October 2018. The actual average price gap after October 2018 is 6.08 yen/kWh between Hokkaido and the East zones, and 0.43 yen/kWh between the West and Kyushu zones. This implies that if the FCFS rule continued, the average price gaps would be lower than the observed price gap under the implicit auction by 2.89 yen/kWh for HH and higher than that by 0.43 yen/kWh for Kanmon. This is an important implication: implicit auction does not necessarily reduce the price gap because it not only has a trade effect, but because it also has a counteracting volume effect.

The actual average trade quantity under the FCFS is 38.26 MW for HH and 31.00 MW for Kanmon. The counterfactual average trade quantity after October 2018 is 23.83 MW for HH and 436.51 MW for Kanmon. The actual average trade quantity after implicit auction increased to 123.55 MW for HH and 1967.56 MW for Kanmon, respectively. Thus, I estimate that implicit auction increased trade quantity on average by 99.72 MW for HH and by 1531.05 MW for Kanmon, relative to the FCFS.

Finally, using Equation (2), I estimate the implicit auction produced economic gain of 4.07 billion yen ($ 40 million) across HH and 6.36 billion yen ($63 million) across Kanmon after the six-month implementation. The larger gain in Kanmon indicates that it has a larger GAIC. The total annual welfare gain was about 20.86 billion yen (USD 208 million). This

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welfare gain was much larger than the one-time implementation cost of the implicit auction, which, according to a member of the JEPX, is 20–30 million yen.

Table 7. Welfare impact of trade under FCFS and implicit auction

	HH	**Kanmon**
Price gap between zones (yen/kWh)		
Autarky price gap (counterfactual)		
FCFS	3.93	0.96
Implicit auction	9.52	1.39
FCFS		
Pre 2018/10 (actual)	3.11	0.63
Post 2018/10 (counterfactual)	3.19	0.86
Implicit auction		
Post 2018/10 (actual)	6.08	0.43
Difference between implicit auction and FCFS	Δ2.89	Δ-0.43
Trade Quantity (MWh)		
FCFS		
Pre 2018/10 (actual)	38.26	31.00
Post 2018/10 (counterfactual)	23.83	436.51
Implicit auction		
Post 2018/10 (actual)	123.55	1967.56
Difference between implicit auction and FCFS	Δ99.72	Δ1531.05
Gain from trade (billion yen/ 6 months)		
Difference between implicit auction and FCFS (Billion yen/ 5 months)	4.07	6.36

9. CONCLUSION

This study examines one of the market based congestion management methods of interconnector transmission capacity: the implicit auction. Implicit auction has two conflicting effects on the day-ahead market, the trade and volume effects. The trade effect arises from the increased trade through interconnectors. Implicit auction increases the interconnector capacity allocated in the day-ahead market and allows for more trade between the export and import zones. The volume effect implies that the implicit auction

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increases the number of electricity bids in the day-ahead market by prohibiting incumbents from reserving interconnector capacity for their bilateral contracts. To simulate market outcomes without implicit auction, counterfactual market outcomes are predicted by comparing linear regression with machine learning methods. I find that the RF generally performs better than linear regression. The welfare impact of the implicit auction is estimated to be approximately $ 40 million for HH and $ 60 million for the Kanmon interconnector after six months of implementation. The annual national production cost savings owing to the trade effect were approximately $ 200 million.

This study highlights the advantages of implicit auction relative to FCFS. Implicit auction not only prevents incumbents from exercising vertical market power, but also promotes efficient resource allocation in the day-ahead market. However, it is important to note that congestion persists after the implicit auction, particularly at the HH interconnector. It reflects the fact that there is a volume effect and the interconnector capacity of the HH was only 0.6 GW by the end of February 2019. With an increase in the amount of variable renewable energy generation, additional investment in interconnectors further increases the benefits of the implicit auction.

There are two limitations to this study due to the lack of available data. First, there is neither interconnector capacity reservation data by company, nor bidding information by company in the day-ahead market. Therefore, it is not possible to verify further whether incumbents exercise market power by using PTR. Second, I cannot estimate the effect of the implicit auction on the reduction of CO2 emission, because transmission system operators only publish aggregate fossil fuel generation data. Implicit auctions may reduce CO2 emissions by decreasing generation from oil and gas-fired power plants that have high marginal costs in an import zone while increasing the CO2 emission by dispatching cheaper coal-fired power plants more frequently in a export zone.

References

Athey, S. and Guido W.I. (2019). “Machine Learning Methods That Economists Should Know About.” Annual Review of Economics 11 (August): 685–725. https://doi.org/10.1146/annurev-economics-080217-053433.

Brunekreeft, G., Karsten N. and David,N. (2005). “Electricity Transmission: An Overview of the Current Debate.” Utilities Policy 13 (2 SPEC. ISS.): 73–93. https://doi.org/10.1016/j.jup.2004.12.002.

Electronic copy available at: https://ssrn.com/abstract=4416730

Bunn, D. and Georg, Z.. (2010). “Inefficient Arbitrage in Inter-Regional Electricity Transmission.” Journal of Regulatory Economics 37 (3): 243–65. https://doi.org/10.1007/s11149-009-9104-5.

Bushnell, J. (1999). “Transmission Rights and Market Power.” The Electricity Journal 12 (8): 77–85. https://doi.org/10.1016/s1040-6190(99)00074-3.

Creti, A., Eileen F., and Elena F. (2010). “Integration of Electricity Markets in Europe: Relevant Issues for Italy.” Energy Policy 38 (11): 6966–76. https://doi.org/10.1016/j.enpol.2010.07.013.

Ehrenmann, A. and Karsten, N. (2009). “A Comparison of Electricity Market Designs in Networks.” Operations Research 57 (2): 274–86. https://doi.org/10.1287/opre.1080.0624.

ETSO, (2004). “An Overview of Current Cross-border Congestion Management Methods in Europe.” https://eepublicdownloads.entsoe.eu/clean-documents/pre2015/publications/etso/Congestion_Management/Current_CM_methods_final_20040908.pdf

Füss, R., Steffen, M. and Marcel, P. (2020). “Electricity Market Coupling in Europe: Status Quo and Future Challenges.” In Handbook of Energy Finance: Theories, Practices and Simulations, 93–120. World Scientific Publishing Co. https://doi.org/10.1142/9789813278387_0005.

Gilbert, R., Karsten N. and David, N. (2004). “Allocating Transmission to Mitigate Market Power in Electricity Networks.” The RAND Journal of Economics 35 (4): 691. https://doi.org/10.2307/1593768.

Gugler, K., Adhurim H. and Mario, L. (2018). “Integration of European Electricity Markets: Evidence from Spot Prices.” Energy Journal 39 (Special Issue 2): 41–66. https://doi.org/10.5547/01956574.39.SI2.kgug.

Joskow, P. L. and Jean, T. (2000). “Transmission Rights and Market Power on Electric Power Networks.” The RAND Journal of Economics 31 (3): 450. https://doi.org/10.2307/2600996.

Keppler, J.H., Sébastien, P. and Yannick, L.P. (2016). “The Impacts of Variable Renewable Production and Market Coupling on the Convergence of French and German Electricity Prices.” The Energy Journal 37 (3). https://doi.org/10.5547/01956574.37.3.jkep.

Electronic copy available at: https://ssrn.com/abstract=4416730

Li, L., Kevin, J., Afshin, R. and Ameet, T. (2018). “Hyperband: A Novel Bandit-Based Approach to Hyperparameter Optimization.” Journal of Machine Learning Research 18: 1–52. http://jmlr.org/papers/v18/16-558.html.

Mansur, E. T. and Matthew, W. W. (2012). “Market Organization and Efficiency in Electricity Markets.” Mimeo.

Marmiroli, M., Masahiko, T., Yukitoki, T. and Ryuichi, Y. (2009). “Market Split Based Congestion Management for Networks with Loops.” IEEJ Transactions on Power and Energy 129 (2). https://doi.org/10.1541/ieejpes.129.265.

Mullainathan, S. and Jann, S. (2017). “Machine Learning: An Applied Econometric Approach.” In Journal of Economic Perspectives, 31:87–106. https://doi.org/10.1257/jep.31.2.87.

Newbery, D., Goran, S, and Ivan, V. (2016). “The Benefits of Integrating European Electricity Markets.” Energy Policy 94 (July): 253–63. https://doi.org/10.1016/j.enpol.2016.03.047.

OCCTO. (2017). Guidelines for Transmission and Distribution of Electricity. https://www.occto.or.jp/article/files/shishin170401.pdf

Twomey, P., Richard, G., Karsten, N. and David, N. (2005). “A Review of the Monitoring of Market Power: The Possible Roles of TSOs in Monitoring for Market Power Issues in Congested Transmission Systems.” CMI Working Paper 71.

Electronic copy available at: https://ssrn.com/abstract=4416730